



PennState

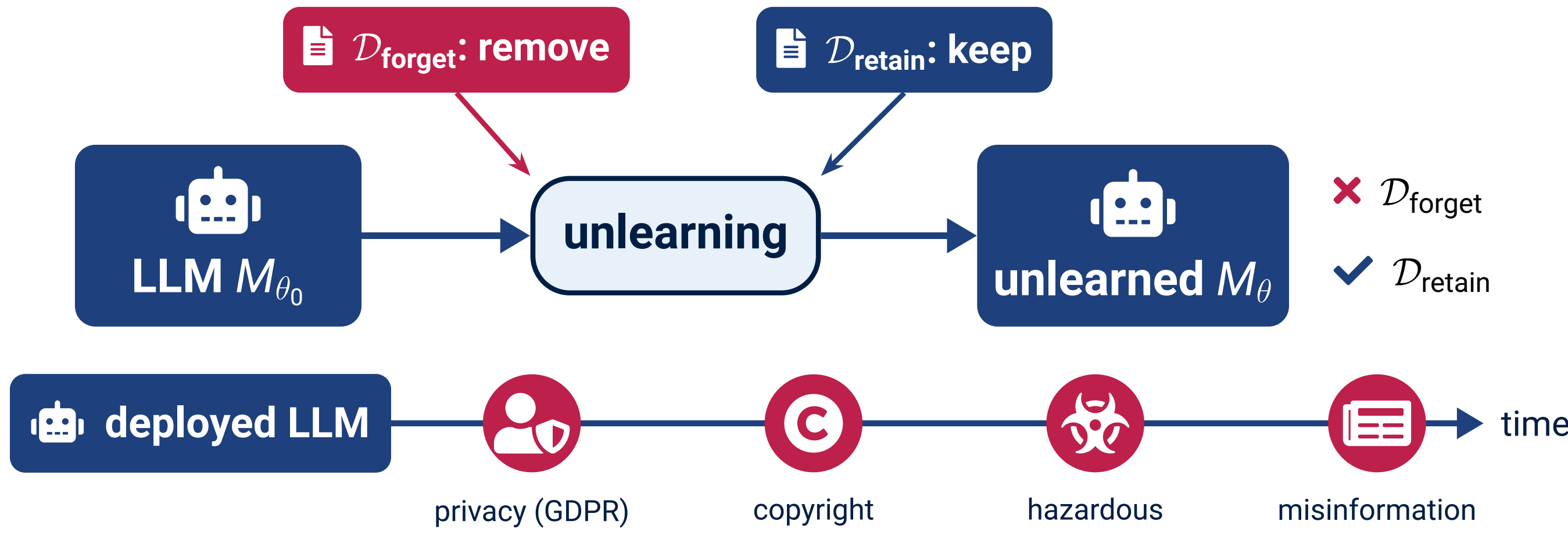
BLADE: Bilevel Low-rank Augmented-Lagrangian Erasure for LLM Unlearning

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Problem Overview



Objectives

Designing an unlearning method that is ...

- Balanced**: Forgets with minimal model degradation
- Scalable**: Stays stable as forget set grows
- Sustainable**: Handles repeated unlearning requests
- Robust**: Resists extraction, jailbreak and membership attacks
- Efficient**: Inexpensive finetuning with smaller memory footprint

BLADE: Method

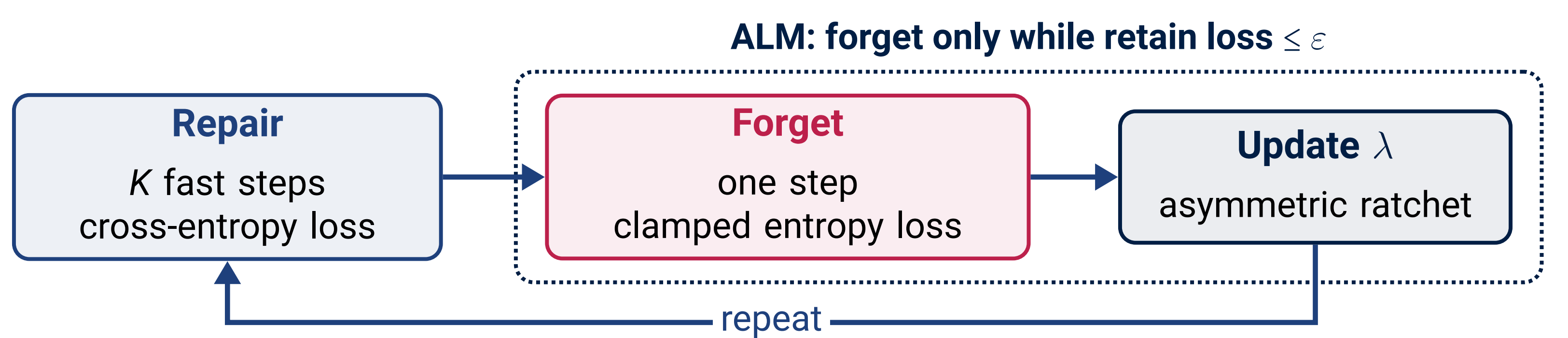
Formulation

$$\min_{\theta} \underbrace{\mathcal{L}_{\text{fgt}}(\theta^*; \mathcal{D}_{\text{forget}})}_{\text{Forget}} \quad \text{s.t.} \quad \underbrace{\mathcal{L}_{\text{CE}}(\theta^*; \mathcal{D}_{\text{retain}}) \leq \varepsilon}_{\text{Retain budget}}$$

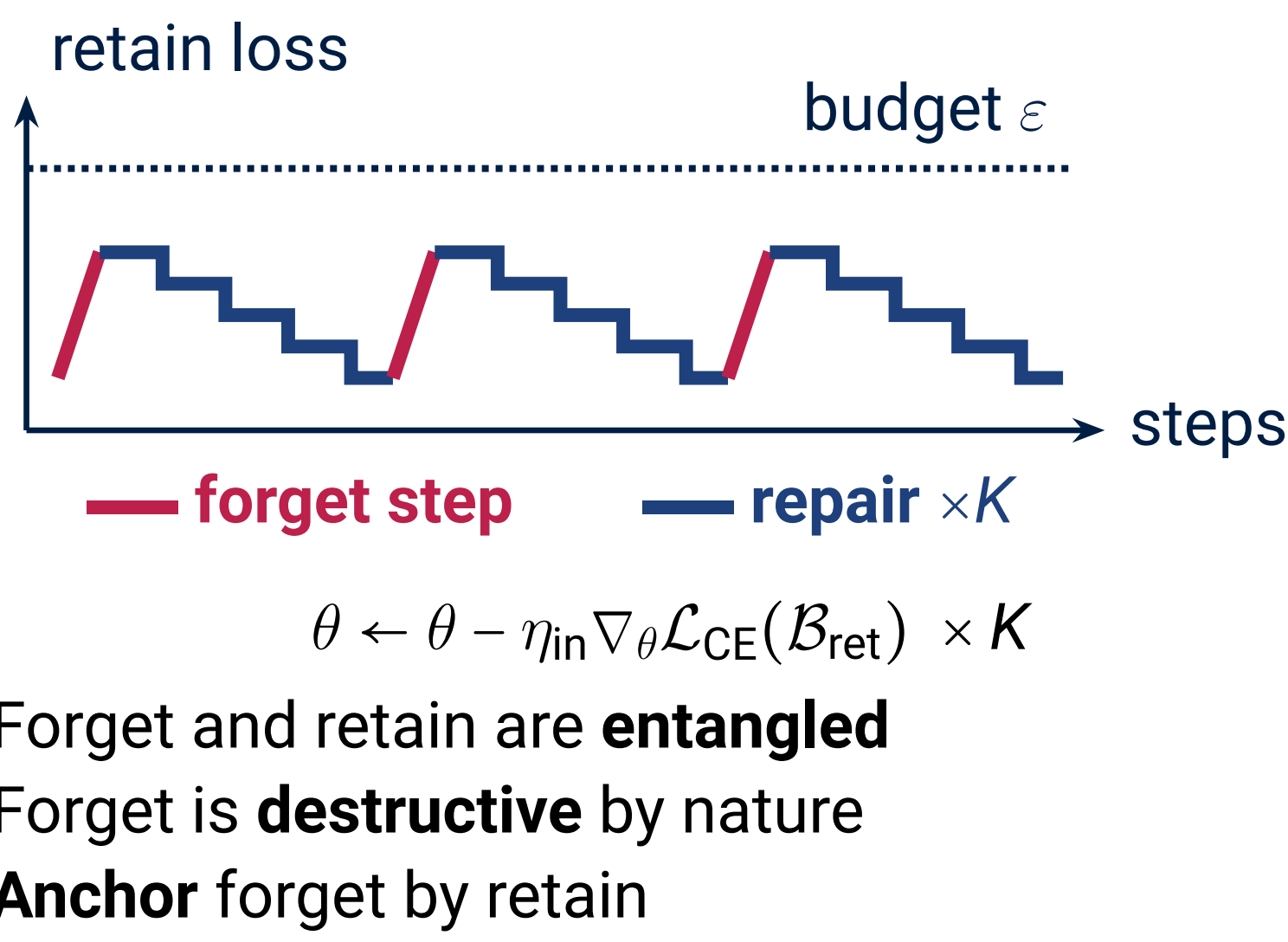
$$\theta^* \approx \arg \min_{\theta} \mathcal{L}_{\text{CE}}(\theta; \mathcal{D}_{\text{retain}}) \quad \underbrace{W' = W_0 + \frac{\alpha}{r} BA}_{\text{LoRA, } W_0 \text{ frozen}}$$

Repair (inner)

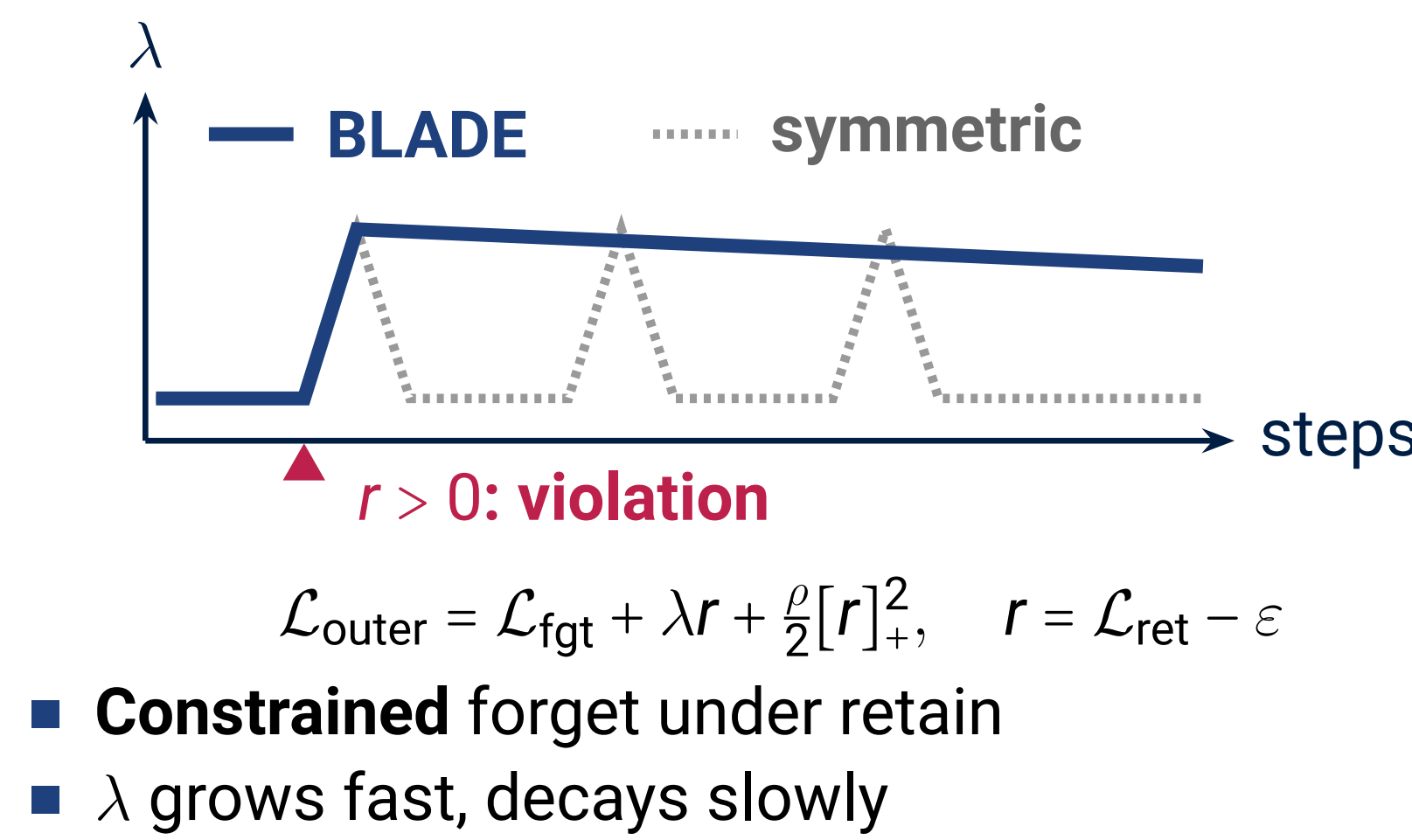
One training step



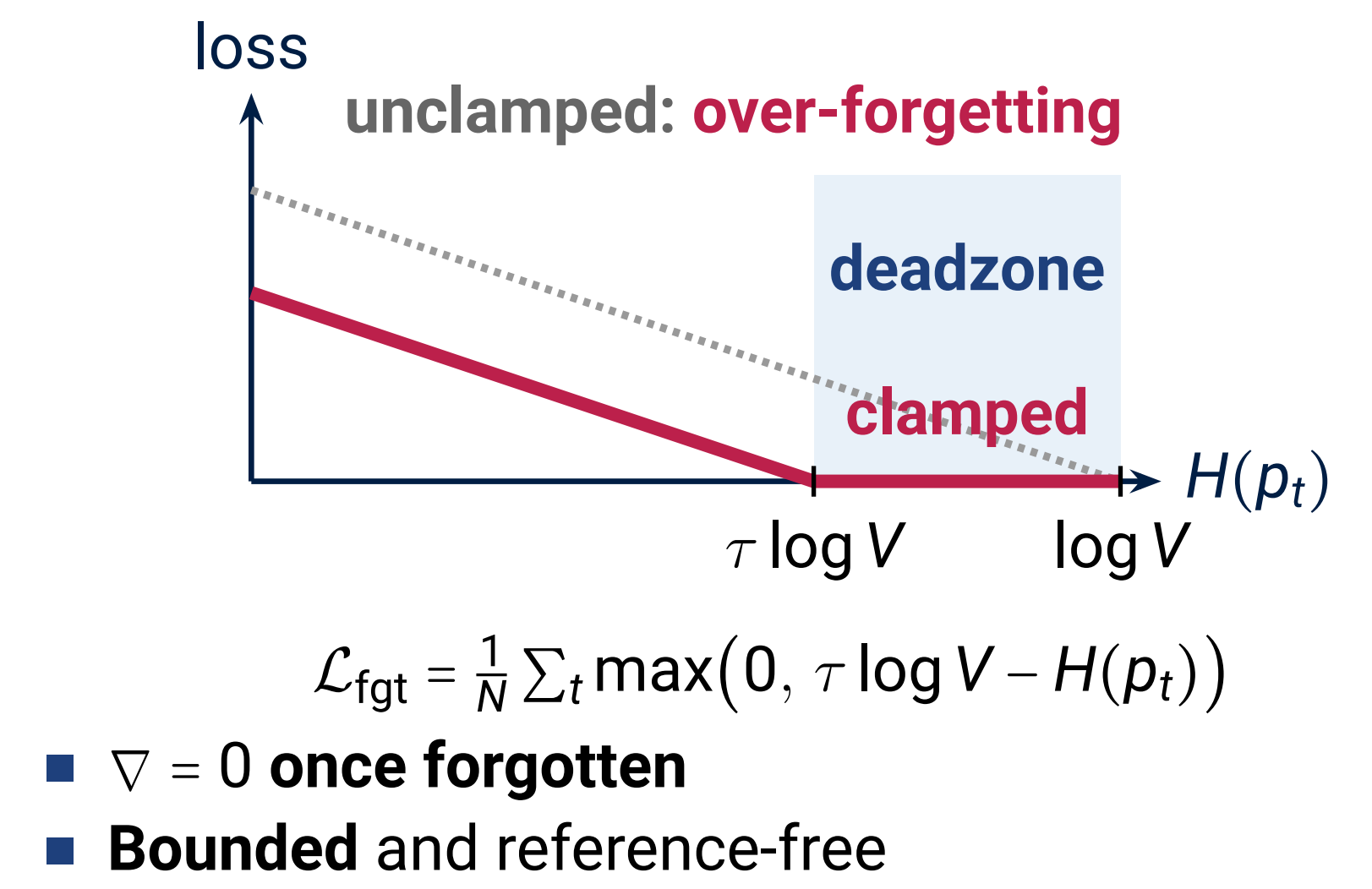
Repair-first principle



Asymmetric ALM



Clamped entropy



Experiments

Setup

Datasets

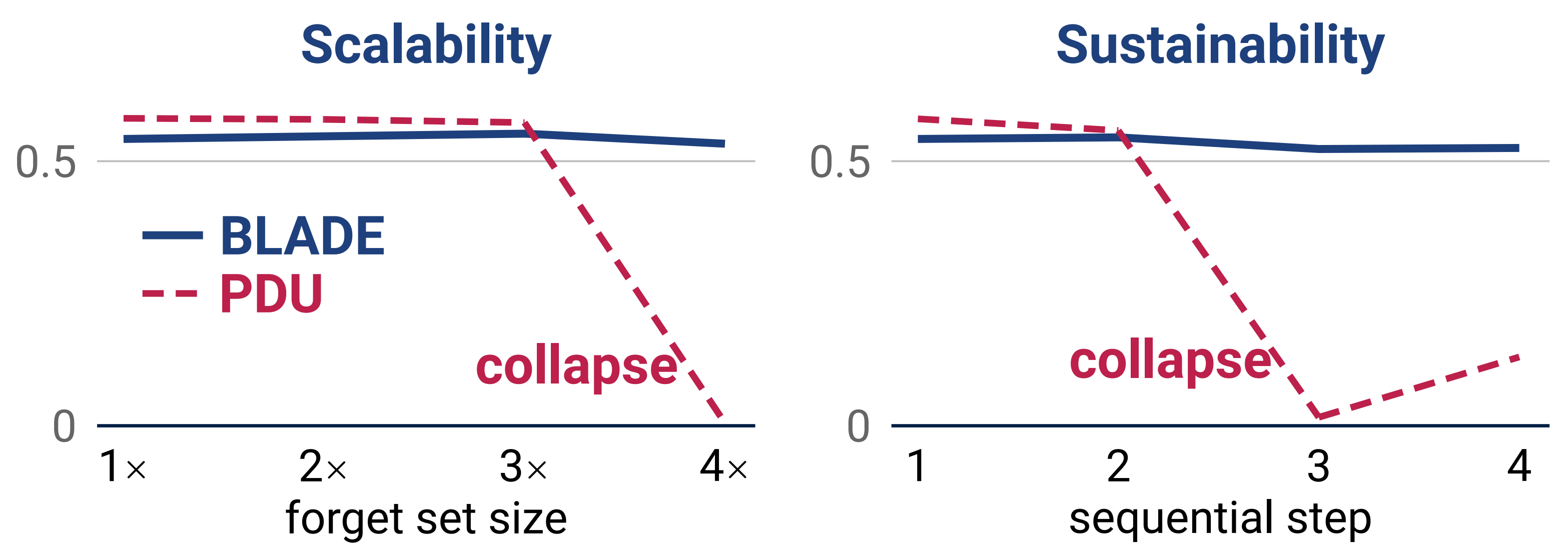
- TOFU**: synthetic authors Llama-3.2 1B/3B
- MUSE**: Books, News Llama-2-7B
- KnowUndo**: copyright, privacy Llama-2-7B-chat

Baselines

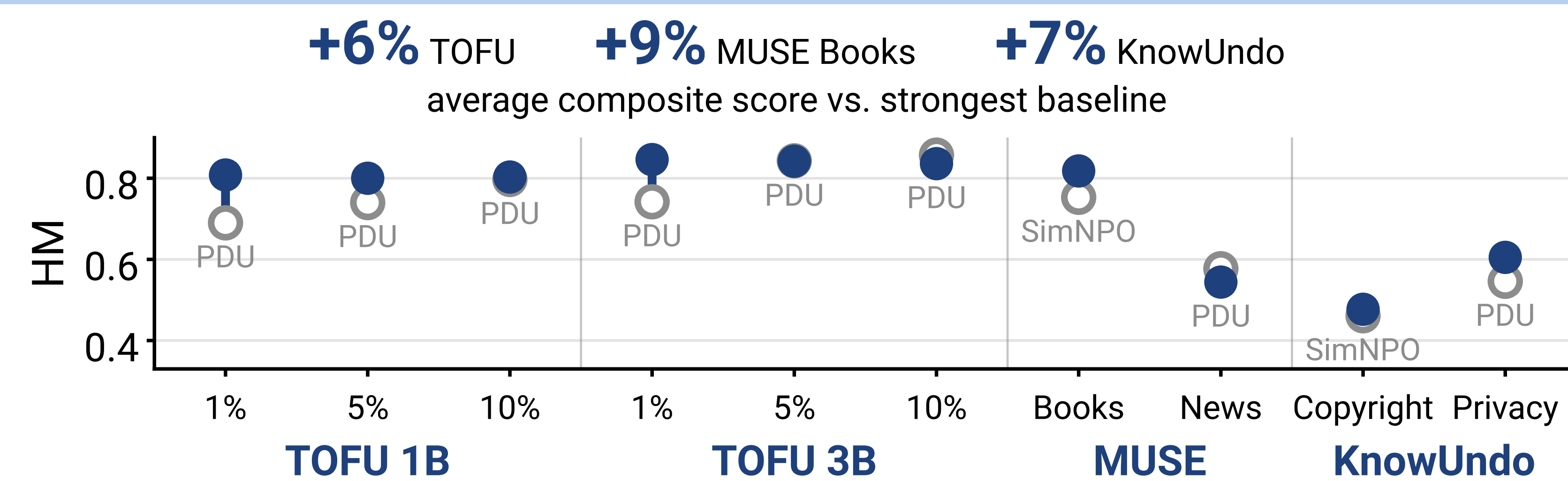
GradAscent³ GradDiff NPO⁵ SimNPO² RMU BLURNPO⁴ PDU¹

Metric: HM of forget & retain axes

Key result: stable under scaling and repeated unlearning



Best on three benchmark families



Robust and efficient

- Extraction**: best adversarial HM
- Jailbreak**: lowest attack success
- MIA**: no leakage
- GCG**: lowest attack success
- Efficient**: 2–5× less GPU memory

1. T. Entesari et al. "Constrained Entropic Unlearning: A Primal-Dual Framework for Large Language Models". 2025.

2. C. Fan et al. "Simplicity Prevails: Rethinking Negative Preference Optimization for LLM Unlearning". 2025.

3. J. Jang et al. "Knowledge Unlearning for Mitigating Privacy Risks in Language Models". 2023.

4. H. Reisizadeh et al. "BLUR: A Bi-Level Optimization Approach for LLM Unlearning". 2026.

5. R. Zhang et al. "Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning". 2024.